

SymPy Bug Card

Bug 30: solveset over the reals treats a principal fractional-power identity as true for negative inputs

Task. Solve the principal fractional-power equation over the real domain.

$$(x^2)^{1/3} = x^{2/3}$$

SymPy's answer.

$$\text{solveset}\left((x^2)^{1/3} = x^{2/3}, x, \mathbb{R}\right) = \mathbb{R}.$$

Correct answer.

$$[0, \infty)$$

Explanation. Substituting $x = -1$, the left side is $((-1)^2)^{1/3} = 1$, while the right side is the principal complex value $(-1)^{2/3} = \exp(2\pi i/3)$. Thus a result containing all negative real numbers includes false solutions.

Likely root cause. The diagnosis traces the call through `unrad`, which cubes both radical terms and collapses the equation to $x^2 = x^2$, and then through `_solve_radical`, which accepts the resulting whole-domain real solution without restoring the lost branch condition. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-030-solveset-over-the-reals-treats-a-principal-fractional-power-/](#)